

**HOLGUIN, Cuba**WMO: **782623**Lat: **20.786N**Lon: **76.315W**Elev: **361**StdP: **14.51**Time Zone: **-5.00 (CUB)**Period: **97-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 59.0       | 62.3 | 51.6                          | 57.4 | 72.7 | 53.6 | 61.9 | 73.9 | 22.8                  | 78.2 | 21.0 | 78.6 | 2.5                   | 50   | 0.342 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 18.0                   | 94.7            | 74.9 | 93.3 | 74.9 | 91.8 | 74.9 | 79.3                | 87.9 | 78.7 | 87.3 | 78.1 | 86.5 | 8.7                  | 90   |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 77.2                            | 143.5 | 82.6 | 76.7 | 140.9 | 82.2 | 75.5 | 135.4 | 81.2 | 43.0          | 88.6 | 42.2 | 87.3 | 41.6 | 86.9 | 91.4              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |             |             | Extreme Annual Temperature |             |                    |            | n-Year Return Period Values of Extreme Temperature |             |             |             |             |             |             |             |    |    |
|-------------------|-------------|-------------|----------------------------|-------------|--------------------|------------|----------------------------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|----|----|
|                   |             |             | Mean                       |             | Standard Deviation |            | n=5 years                                          |             | n=10 years  |             | n=20 years  |             | n=50 years  |             |    |    |
| 1%                | 2.5%        | 5%          | Min                        | Max         | Min                | Max        | Min                                                | Max         | Min         | Max         | Min         | Max         | Min         | Max         | DB | WB |
| (a)               | (b)         | (c)         | (e)                        | (f)         | (g)                | (h)        | (i)                                                | (j)         | (k)         | (l)         | (m)         | (n)         | (o)         | (p)         |    |    |
| (4) <b>20.6</b>   | <b>18.6</b> | <b>16.7</b> | <b>53.7</b>                | <b>96.7</b> | <b>3.5</b>         | <b>1.1</b> | <b>51.2</b>                                        | <b>97.5</b> | <b>49.2</b> | <b>98.1</b> | <b>47.2</b> | <b>98.7</b> | <b>44.7</b> | <b>99.5</b> |    |    |
| (5)               |             |             | <b>51.8</b>                | <b>81.6</b> | <b>3.2</b>         | <b>2.3</b> | <b>49.5</b>                                        | <b>83.2</b> | <b>47.7</b> | <b>84.6</b> | <b>45.9</b> | <b>85.8</b> | <b>43.6</b> | <b>87.5</b> |    |    |

**Monthly Climatic Design Conditions**

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 79.2   | 74.2  | 75.6  | 76.8  | 78.7  | 80.7  | 82.5  | 83.3  | 83.2  | 82.2  | 80.3  | 76.9  | 75.6 |
|                                                                               | DBStd       | 4.12   | 3.52  | 3.41  | 3.54  | 2.76  | 2.32  | 2.04  | 1.59  | 1.73  | 1.59  | 2.21  | 2.99  | 3.39 |
|                                                                               | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | HDD65       | 2      | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1    |
|                                                                               | CDD50       | 10653  | 752   | 717   | 832   | 862   | 950   | 974   | 1031  | 1028  | 965   | 940   | 808   | 794  |
|                                                                               | CDD65       | 5179   | 287   | 297   | 367   | 412   | 485   | 524   | 566   | 563   | 515   | 475   | 358   | 330  |
|                                                                               | CDH74       | 45026  | 1837  | 2236  | 2932  | 3532  | 4281  | 5154  | 5891  | 5756  | 4762  | 3945  | 2520  | 2180 |
|                                                                               | CDH80       | 18272  | 532   | 783   | 1131  | 1445  | 1808  | 2253  | 2657  | 2629  | 2126  | 1533  | 745   | 630  |
| Wind                                                                          | WSAvg       | 7.6    | 8.1   | 8.6   | 8.8   | 8.6   | 7.8   | 7.2   | 8.0   | 7.6   | 6.0   | 6.2   | 7.1   | 7.7  |
| Precipitation                                                                 | PrecAvg     | 43.2   | 1.5   | 1.4   | 1.6   | 2.5   | 5.6   | 5.2   | 4.2   | 5.0   | 5.7   | 5.5   | 3.1   | 1.8  |
|                                                                               | PrecMax     | 54.8   | 4.3   | 3.6   | 4.5   | 4.4   | 10.8  | 9.7   | 6.8   | 7.5   | 10.6  | 12.4  | 8.4   | 7.0  |
|                                                                               | PrecMin     | 29.0   | 0.4   | 0.4   | 0.6   | 1.2   | 2.9   | 2.7   | 2.2   | 3.1   | 3.0   | 2.0   | 0.4   | 0.6  |
|                                                                               | PrecStd     | 5.3    | 0.8   | 0.9   | 0.8   | 0.9   | 2.0   | 1.7   | 1.1   | 1.1   | 1.5   | 2.5   | 1.9   | 1.2  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 88.2  | 91.3  | 93.3  | 93.4  | 94.7  | 95.2  | 95.3  | 95.4  | 94.8  | 92.8  | 90.5  | 89.4 |
|                                                                               |             | MCWB   | 69.3  | 69.1  | 70.1  | 71.1  | 74.0  | 75.0  | 75.4  | 75.4  | 74.6  | 74.9  | 74.3  | 71.7 |
|                                                                               | 2%          | DB     | 87.1  | 89.4  | 91.1  | 91.4  | 92.8  | 93.3  | 93.6  | 93.6  | 93.1  | 91.2  | 88.1  | 87.4 |
|                                                                               |             | MCWB   | 69.7  | 68.8  | 70.5  | 71.9  | 73.4  | 75.5  | 75.3  | 75.8  | 75.5  | 75.5  | 74.5  | 72.4 |
|                                                                               | 5%          | DB     | 85.5  | 87.5  | 89.2  | 89.6  | 91.1  | 91.6  | 92.9  | 92.9  | 91.5  | 89.6  | 86.2  | 85.8 |
|                                                                               |             | MCWB   | 70.2  | 69.4  | 70.3  | 71.4  | 73.6  | 75.6  | 75.4  | 75.8  | 75.9  | 75.7  | 74.1  | 72.5 |
|                                                                               | 10%         | DB     | 82.8  | 85.6  | 86.3  | 87.8  | 89.3  | 89.9  | 91.2  | 91.3  | 89.9  | 87.9  | 84.3  | 84.0 |
|                                                                               |             | MCWB   | 70.0  | 69.9  | 70.0  | 71.0  | 73.5  | 75.7  | 75.6  | 75.9  | 76.1  | 75.9  | 73.3  | 72.0 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 74.5  | 74.4  | 76.8  | 76.6  | 79.1  | 80.0  | 79.7  | 80.1  | 80.0  | 79.2  | 78.1  | 76.3 |
|                                                                               |             | MCDB   | 82.0  | 83.1  | 86.4  | 86.5  | 88.5  | 88.7  | 89.1  | 89.1  | 88.4  | 86.4  | 83.9  | 83.3 |
|                                                                               | 2%          | WB     | 73.1  | 73.0  | 73.9  | 75.1  | 77.6  | 78.8  | 78.6  | 79.0  | 79.0  | 78.3  | 76.7  | 75.0 |
|                                                                               |             | MCDB   | 80.2  | 81.2  | 82.8  | 83.7  | 87.1  | 87.7  | 87.9  | 87.8  | 87.3  | 85.7  | 83.4  | 82.4 |
|                                                                               | 5%          | WB     | 72.4  | 72.2  | 72.7  | 74.0  | 76.5  | 77.8  | 77.8  | 78.2  | 78.2  | 77.4  | 75.8  | 74.3 |
|                                                                               |             | MCDB   | 79.4  | 80.3  | 81.4  | 82.9  | 85.3  | 86.3  | 86.8  | 86.7  | 86.4  | 85.1  | 82.5  | 81.2 |
|                                                                               | 10%         | WB     | 71.6  | 71.3  | 71.8  | 73.0  | 75.6  | 76.9  | 77.0  | 77.4  | 77.4  | 76.6  | 74.9  | 73.4 |
|                                                                               |             | MCDB   | 78.4  | 79.3  | 80.4  | 81.8  | 83.6  | 85.1  | 85.7  | 85.7  | 85.5  | 84.2  | 81.4  | 79.8 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 17.6  | 19.2  | 19.8  | 19.4  | 18.3  | 17.3  | 18.0  | 18.5  | 18.4  | 16.4  | 15.9  | 16.5 |
|                                                                               |             | MCDBR  | 20.5  | 22.1  | 22.4  | 21.4  | 20.4  | 18.8  | 19.6  | 19.9  | 19.6  | 18.2  | 17.5  | 18.3 |
|                                                                               |             | MCWBR  | 7.2   | 6.9   | 6.5   | 6.0   | 6.1   | 6.0   | 5.7   | 6.1   | 6.5   | 6.4   | 6.7   | 6.8  |
|                                                                               | 5% WB       | MCDBR  | 16.9  | 19.1  | 19.8  | 19.5  | 17.5  | 17.6  | 18.2  | 18.3  | 18.2  | 16.5  | 15.1  | 16.3 |
|                                                                               | MCWBR       | 6.7    | 6.7   | 6.6   | 6.3   | 6.5   | 6.5   | 6.3   | 6.6   | 6.8   | 6.6   | 6.5   | 6.7   |      |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.366  | 0.372 | 0.380 | 0.419 | 0.440 | 0.476 | 0.483 | 0.455 | 0.423 | 0.404 | 0.384 | 0.382 |      |
|                                                                               | taud        | 2.519  | 2.495 | 2.475 | 2.366 | 2.346 | 2.237 | 2.203 | 2.311 | 2.421 | 2.477 | 2.491 | 2.488 |      |
|                                                                               | Ebn at Noon | 288    | 293   | 294   | 282   | 272   | 261   | 259   | 268   | 277   | 279   | 280   | 279   |      |
|                                                                               | Edh at Noon | 31     | 34    | 36    | 40    | 41    | 45    | 46    | 42    | 37    | 34    | 32    | 31    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 1326   | 1596  | 1803  | 1930  | 1879  | 1913  | 2005  | 1938  | 1782  | 1514  | 1346  | 1259  |      |
|                                                                               | RadStd      | 92     | 103   | 96    | 112   | 110   | 97    | 88    | 49    | 100   | 108   | 92    | 85    |      |

**Historical Trends**

|      |                        | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|------------------------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                        | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only           | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (3 neighbors) | N/A     | N/A    | +2.99   | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page